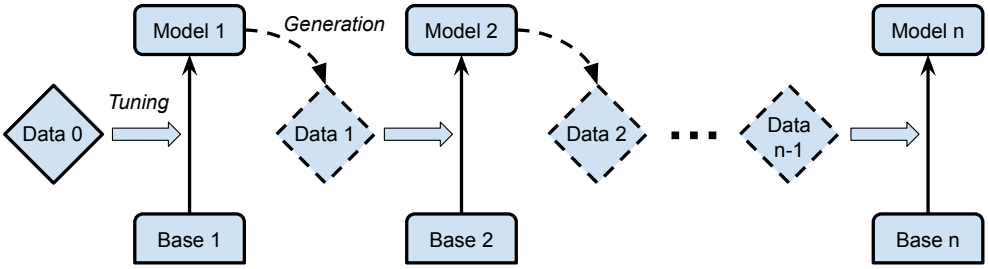

Figure 1: Our recursive tuning-generation process. Beginning with authentic, human-curated Data (0), Base (1) model undergoes finetuning to develop Model (1), which is the first model subject to our language diversity research. Subsequently, we use Model (1) to create synthetic Data (1) to train a successor Model (2) of the next generation, on the basis of Base (2) model. The process continues for n iterations. Base (1), Base (2), ..., Base (n) follow the same model architecture but are independently initialized instances.

texts can be grammatically sound and contextually coherent yet still score high in perplexity if they include unusual or creative language not included in the model's training data (Basu et al., 2021). In our study, a model with lower perplexity is not deemed superior by default. Our aim is to ensure that the perplexity remains within a reasonable limit, producing texts of sufficient quality for our linguistic diversity evaluation.

3.3 Linguistic Diversity Metrics

We approach the evaluation of linguistic diversity from three different perspectives: lexical diversity, semantic diversity and syntactic diversity.

3.3.1 Lexical Diversity

Lexical diversity metrics are used to measure the variety of words used in a text, which is contended to mirror the extent of vocabulary possessed by a writer or speaker. We believe a degenerated language model, which presumably has a smaller vocabulary, will use a narrower variety of lexical items than non-degenerated language models. We select different metrics operating at different levels of textual granularity: word, n-gram, and sentence.

Type-Token Ratio (TTR) (Johnson, 1944; Templin, 1957), the most well-known metric, which is calculated as the number of unique words (types) t divided by the number of running words (tokens) c, i.e., TTR $= t/c$. This metric was used to study the language development in child language research, a low value is probably indicative of a language-specific deficiency (Miller, 1981). The length of a text inherently skews vanilla TTR values, with longer texts generally yielding lower TTR scores due to an inexorably decreased occurrence of unique novel words (drawn from a limited vocabulary) as the text lengthens (Richards, 1987). Following common practice (Shaib et al., 2024), we truncate all texts to a fixed length before computing the TTR [2].

Distinct-n (Li et al., 2016), which equals the proportion between the number of unique n-grams and total number of n-grams in tested text (Xing et al., 2017). This metric is originally introduced in the context of enhancing the response diversity of conversational agents, which frequently produce safe and fluent but dull and uninformative generic responses at time (e.g., *I don't know*) (Han et al., 2022). Similar to naive TTR, distinct-n varies as a function of text length, so we report the results at fixed sizes for $n = 2$ and $n = 3$ (distinct-1 is equivalent to TTR).

Self-BLEU (Zhu et al., 2018), a recently developed method for evaluating the diversity of synthetic text. This method assesses the similarity between one sentence and the rest in a group of generated sentences. It treats one sentence as the hypothesis and the others as references to calculate BLEU scores (Papineni et al., 2002). The final Self-BLEU score averages these BLEU scores across all generated sentences. We report $1 - \texttt{Self-BLEU}$, so a higher value reflects richer diversity of the generation (Palumbo et al., 2020).

3.3.2 Semantic Diversity

According to recent studies (Tevet and Berant, 2021; Stasaski and Hearst, 2022), the above lexical-level metrics often fail to capture semantic diversity, since texts including similar words can have different semantics and texts with different words can have similar semantics (Yarats and Lewis, 2018). We tackle this problem by transforming sentences

[2] Details on the truncation lengths can be found in Appendix B.

into semantically meaningful sentence embeddings using Sentence-BERT (Reimers and Gurevych, 2019). We quantify semantic diversity as the dispersion of sentence embeddings over the semantic space. The dispersion is measured by the average pairwise cosine-distance of all embedding vectors (`Div_sem`).

3.3.3 Syntactic Diversity

The significance of syntactic diversity is often underestimated in NLP, despite its importance. For language learners (as well as language models), exposure to a wide range of syntactic structures is beneficial for developing a more comprehensive understanding of the language (Aggarwal et al., 2022). Moreover, a range of syntactic forms enhances expressiveness and subtlety in writing, influencing the style and tone of a text (Edwards and Bastiaanse, 1998). While linguistic and language acquisition research (Clercq and Housen, 2017) has explored this aspect, these studies typically rely on manual annotation of features, a process that can be costly and prone to human error.

We introduce the first graph-based metric to quantify syntactic diversity. We use a neural parser (Qi et al., 2020) to construct dependency trees from sentences, following the universal dependencies formalism. These trees are then transformed into graph representations, with nodes representing the words and edges indicating the dependency relationships between them. Subsequently, we employ the Weisfeiler-Lehman graph kernel (Shervashidze et al., 2011; Siglidis et al., 2020) to map these graphs into a vector space. This kernel, rooted in the Weisfeiler-Lehman isomorphism test, effectively positions graphs that are structurally alike closer to each other in the embedding space. To assess syntactic diversity, we calculate it similarly to semantic diversity, using the average pairwise distance (`Div_syn`).

4 Experiments and Results

We conduct our experiments on three generative tasks, as introduced in Section 3.1, with decreasing degrees of constraint and increasing degrees of creativity: abstractive news summarization, scientific abstract generation, and story generation.

4.1 Experimental Setup

For each task, we simulate 6 iterations of the recursive training chain, i.e., $n = 6$ in Figure 1. Following previous work (Shumailov et al., 2023), we select OPT (Zhang et al., 2022) as our base model, and each iteration begins with a new instance of the base model. Different from Shumailov et al. (2023), we use OPT-350M instead of OPT-125M to maintain higher generation quality over iterations, avoiding excessive noise. Model (1) is finetuned on Data (0)—the training set of the finetuning task—which is human-authored. From Model (2) to Model (6), they are finetuned on synthetic Data ($n-1$) generated by their predecessor Model ($n-1$). We go through all of the original training examples in Data (0) to produce a comparable synthetic dataset of the same number of samples. The models are finetuned for 5 epochs using the AdamW optimizer (Loshchilov and Hutter, 2019) on a cluster of two NVIDIA RTX A6000 GPUs.

In the following, we explain each of the three tasks in detail.

Task1: Abstractive News Summarization

For abstractive news summarization, we use the XL-SUM (Hasan et al., 2021), one of the most recently proposed datasets. In comparison to the other prominent news summarization datasets, XL-SUM is more abstractive than CNN/DailyMail (Hermann et al., 2015; Cheng and Lapata, 2016) and more factual than XSUM (Narayan et al., 2018; Guo et al., 2022). It is also larger in scale, consisting of 306,522 samples in the training set and 11,535 samples in the test set. The average length of the news articles is 386 tokens, while the average length of the summaries is 28 tokens. This is a generation task with "low entropy" since there is abundant context and the content is restricted.

Task2: Scientific Abstract Generation

For scientific abstract generation, we parse the bibliography database (BibTeX) file with abstracts of ACL Anthology[3] in June, 2023. ACL Anthology hosts papers published at computational linguistics or natural language processing venues since 1965. We split the bibliography entries into the training and the test set, resulting in 40,512 samples for train and 2,132 samples for test. We use the title of the paper and the first sentence of the abstract as the prompt, asking the model to finish generating the rest of the abstract. The prompt (title + first sentence) is 42 tokens long on average, while the mean of the full abstract length is 145 tokens. This is a task of "medium entropy", the provided title

[3] https://aclanthology.org

Prompt: The title of the paper is: CASIA's System for IWSLT 2020 Open Domain Translation.The abstract of the paper is: This paper describes the CASIA's system for the IWSLT 2020 open domain translation task.
Human: This year we participate in both Chinese→Japanese and Japanese→Chinese translation tasks. Our system is neural machine translation system based on Transformer model. We augment the training data with knowledge distillation and back translation to improve the translation performance. Domain data classification and weighted domain model ensemble are introduced to generate the final translation result. We compare and analyze the performance on development data with different model settings and different data processing techniques..
Iteration 1: We participated in all language directions, namely Chinese ↔ English, Japanese↔ English and German ↔ English. Our system mainly focuses on data selection, data preprocessing and data filtering to improve the quality of the generated translations..
Iteration 3: We participated in all three language pairs, namely Chinese↔English, Japanese→Chinese and German↔French. Our systems are based on Transformer architecture with a simple but effective method to improve the translation quality by reusing monolingual data from other languages.. The source code can be obtained from `https://github.com/csebuetnlp/acl2021/opendomaintranslate`..The blind test set will be released upon the acceptance of the report..The full results can also be found at `https://competitions.codalab.org/competitions/26153`.. The leaderboard can be found at `https://competitions.codalab.org/page/view/11650`..The dataset used in this work has been released under CC BY 4.0 license..The experimental results show that our proposed methods significantly outperform the baseline models..
Iteration 5: We participated in all three language pairs, namely Chinese↔English (Chinese→German), Japanese→Czech and German↔French.. Our approach relies on a Transformer model trained with in-domain data only; it also incorporates a novel method to improve the quality by leveraging parallel corpora from other domains such as TED Talks. The code used in this work has been released under CC BY 4.0 license at `https://github.com/czyssrs/ibm_nlp2020/`. For further information about our systems please see `http://www.csse.unige.edu/research/demo/`.. The full text of this report can be obtained via `https://competitions.codalab.org/page/view/11153`.. The codes for our experiments are available online at `https://codebenchmark.io/`, too..... Finally, we provide some insights into how our systems perform when translating into low resource conditions..... The source code for our systems is publicly available through `https://github.com/czeneva/acl2022tasks2/`.

Table 1: An example of model outputs over iterations. Since diversity is a holistic concept observed among the entire set of documents, it is non evident to demonstrate the loss of diversity through a single example. However, for the scientific abstract generation task in this example, there are some obvious patterns. Abstracts found in the ACL anthology often contain URLs linking to data, code or demos. After a few iterations of training on synthetic text, the model picked up this high probability event and starts to generate such links repeatedly in an unnatural manner.

already lays out the general idea of the paper and the first sentence provides a fair amount of context.

Task3: Story Generation

For story generation, we use the WritingPrompts dataset (Fan et al., 2018). It is made up of human written stories paired with writing prompts from Reddit's WritingPrompts forum. There are 272,600 samples in the training set and 15,138 samples in the test set. The writing prompts consist of 30 tokens on average and the resulting stories have a mean of 389 tokens. The prompts are generally short and in most cases do not contain a plot (i.e. narrative structure), making this a "high-entropy" generation task with limited context.

Decoding Strategy.

We use a combination of nucleus sampling (p) and temperature sampling (τ) to achieve nuanced control over the language model's outputs (Holtzman et al., 2020). Nucleus sampling, also known as top-p sampling, is used to generate text by selecting the most probable words from a distribution of words. It ensures that the cumulative probability of the chosen words exceeds a certain threshold (p). Higher values of p lead to more deterministic text. Temperature sampling involves dividing the output logits by a temperature parameter (τ) before sampling from the distribution. Higher values of τ make the distribution more uniform and increase randomness.

We adapt the specific parameters to the characteristic of each task (Amini et al., 2023). For news summarization, we emphasize precision and set $p = 0.1$, $\tau = 0.3$. For story generation, we care more about creativity and set $p = 0.9$, $\tau = 0.7$. For scientific abstract generation, we search something in between and set $p = 0.5$, $\tau = 0.5$. The `max_new_tokens` value is chosen according to the length of human-written references for each task: 50 for news summarization, 500 for story generation and 300 for scientific abstract generation. While the decoding strategy has influence over diversity metrics, it is not the determinant factor (Giulianelli et al., 2023). We aim to draw generalizable conclusions by experimenting across three distinct sets of decoding strategies.